



Ownership and capital structure in Latin America[☆]

Jacelly Céspedes^a, Maximiliano González^{b,*}, Carlos A. Molina^a

^a Instituto de Estudios Superiores de Administración (IESA), Ave IESA, San Bernardino, Caracas 1010, DF, Venezuela

^b Universidad de los Andes (UNIANDES), Calle 21, No. 1-20, Edif. JMSD, Ofic. 914 Bogotá, Colombia

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ABSTRACT

This study evaluates the capital-structure determinants of Latin American firms using a comprehensive sample covering seven countries. Firms in the region have debt levels similar to those of U.S. firms, which is puzzling, given that Latin American firms experience relatively lower tax benefits and higher bankruptcy costs. This study argues that ownership-concentrated firms avoid issuing equity because they do not want to share control rights. Latin American firms have high ownership concentration, which creates an ideal setting to study how ownership concentration explains firms' capital structure. Consistent with the control argument, this study finds a positive relation between leverage and ownership concentration, when losing control becomes an issue. Also, the study shows a positive relation between leverage and growth. In addition, the study reports that other determinants that do not proxy for control rights are consistent with previous findings. Firms that are larger, have more tangible assets, and are less profitable are also more leveraged.

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1. Introduction

This study argues that Latin American firms exhibit higher leverage than expected under the trade-off theory of capital structure, because their ownership concentration is significantly higher to those in developed economies. In Latin America, stockholders of firms with highly concentrated ownership prefer to issue debt rather than equity because Latin American shareholders do not want to risk losing control of their companies. As shown in Chong and López-de-Silanes (2007), larger information asymmetries and underdeveloped financial markets also make the issuance of equity less likely in the region.

A recent debate exists on how the benefits and costs of debt determine an equilibrium capital structure for companies in the United States. According to Graham (2000) firms are less leveraged than they should be given the tax benefits of debt. Molina (2005), on the other hand, shows how costs of financial distress measured ex-ante can counterbalance the potential tax benefits that an average firm will obtain when leveraging up.

In Latin America the situation may be quite different. La Porta et al. (1997, 1998, 2000) report that the legal systems firms face differ significantly around the world, in part, because of their home countries' legal origins. They argue that investors are less protected

in French Civil Law countries, compared to countries with Common Law origins. All countries in Latin America have the same legal origin, which is French Civil Law. Chong and López-de-Silanes (2007) show that in Latin America, whose countries offer less investor protection than average French Civil Law countries, investors' expropriation risk is more severe, the cost of capital is higher, firms pay less in dividends, and, in general, the level of financial development in the region is very low.

In terms of the international debate of the optimal capital structure, the situation in Latin America offers interesting insights. First, tax benefits tend to be lower. In Chile, for example, corporate taxes can be deducted from personal income taxes, reducing the corporate tax-shield to as little as zero. In Mexico and Venezuela, required inflation adjustments on corporate financial statements can turn the tax shield negative in some cases. On the other hand, the costs of financial distress are presumably higher in Latin America than in the United States. In Latin America, bankruptcy and financial-distress processes are longer, more bureaucratic, and more costly due in part to a less robust rule of law (La Porta et al., 1997).

If Latin American firms have on average fewer tax benefits and higher costs of financial distress than their U.S. counterparts, then, according to the trade-off theory of capital structure, they should be significantly less leveraged. In addition, the argument by Jensen (1986) that debt could act as an effective mechanism to monitor management does not apply to Latin American firms. In Latin America the typical CEO is, or is related to, one of the firm's largest controlling shareholders. According to Johnston (2004), only two shareholders typically hold more than 50% of a firm's equity.

Latin American firms have lower debt levels for other reasons. Financial markets are less developed and the economies have been

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* Corresponding author.

E-mail addresses: jacelly.cespedes@iesa.edu.ve (J. Céspedes), mgf@adm.uniandes.edu.co (M. González), carlos.molina@iesa.edu.ve (C.A. Molina).

more volatile, leaving firms with fewer financing options. In some countries, this has meant eliminating long-term financing altogether (Chong and López-de-Silanes, 2007). Silva, Majluf and Paredes (2006) offer evidence on how Chilean firms form conglomerates to overcome the lack of alternatives to financing through internal capital markets. Moreover, the social capital managers develop in a network-oriented business environment such as Latin America is another source of financing (Young and Tsai, 2008).

Surprisingly, the study does not find lower levels of leverage for Latin American firms in this study. Instead, when looking at a comprehensive sample of Latin American firms from seven countries, this study finds their book-value leverage ratios to be similar to those reported by Rajan and Zingales (1995) and Flannery and Rangan (2006) for U.S. firms.

Rajan and Zingales (1995) contend that the effect of ownership concentration on capital structure is far from obvious. On one hand, the presence of large shareholders on the board of directors should reduce the extent of agency costs between managers and shareholders and facilitate equity issuances (Lefort and Urzua, 2008). Furthermore, these shareholders would be undiversified, increasing their aversion to debt. On the other hand, if some of the shareholders are banks, they may force the firms to borrow from them. In Latin America the banks are not large shareholders as they are in Japan and Germany. But firm control plays an important role in Latin America given the lack of protection for minority shareholders.

At the aggregate level the results support the idea that ownership structure strongly influences firms' leverage. This study finds a U-shape relation, where at high levels of ownership concentration the relation between leverage and ownership is positive, consistent with the study's previous argument that companies with highly concentrated ownership do not seek equity financing in order to avoid losing control. On the other hand, at low levels of ownership concentration the relation between leverage and ownership is negative. This study maintains that when control is not an issue—the inflection point occurs at the middle of the sample when the first shareholder has at least 45% of ownership—undiversified shareholders act on their aversion to debt.

Latin American firms follow a pecking order of financing alternatives when they need funds. When ownership is dispersed, and given that Latin American investors are not well protected, this study expects that the monitoring role of debt adds some value for those firms. At the country level however, these results are strong only for Brazil and Chile, the two countries with more observations in the sample. With the exception of Mexico, the coefficient signs, although not significant, are as expected for all other countries. This is remarkable given the small number of observations for countries such as Argentina, Colombia, and Venezuela.

The paper also shows how the determinants of capital structure referred to in the literature apply to Latin American firms. Bradley et al. (1984), Titman and Wessels (1988), Kale et al., (1992), Flannery and Rangan (2006), and Hovakimian (2006) are part of the previous empirical literature that use large U.S. samples to study the capital-structure determinants. The literature also explores capital-structure determinants outside the United States. Rajan and Zingales (1995) use data from seven developed economies, and Booth et al. (2001) use data from ten developing economies but only include two Latin American countries: Brazil and Mexico. This study empirically tests if factors that are not proxies for ownership control—such as asset tangibility, size, growth opportunities, and profitability—help to explain the leverage in Latin American firms. This is one of the first regional studies on the determinants of capital structure.

Consistent with the literature, this study finds that leverage is positively related to size and asset tangibility. Larger firms have more access to financing sources, and firms with more tangible assets have more collateral to support higher debt levels. Also consistent with previous literature is the finding that profitability is negatively related

to leverage—further support for the pecking-order theory for Latin American firms.

The study's findings on growth, however, contradict the literature but are consistent with the study's control argument. Titman and Wessels (1988) find that agency costs and suboptimal investments lead growing firms to have less debt. Rajan and Zingales (1995) also refer to suboptimal investments to argue that firms with growth opportunities should use more equity. Contrary to previous literature, this study finds a positive relation between leverage and growth. Growing Latin American firms that need cash avoid equity issuances, because their shareholders do not want to lose control.

The rest of the paper proceeds as follows. Section 2 describes the data set, explains the constructions of the variables, and shows the principal descriptive statistics. Section 3 reports and discusses the results. Finally, Section 4 presents the study's conclusions.

2. Data and research design

The sample includes all Latin American firms with data available in the Economatica database from 1996 to 2005. Economatica has financial data for firms in seven Latin American countries: Argentina, Brazil, Chile, Colombia, Mexico, Peru, and Venezuela. The database includes firms that report their financial statements to local regulatory agencies. It includes quarterly company financial statements and shareholder information, as well as financial and market data. Ownership-concentration data are collected from Economatica for Brazil, Chile, Colombia, Peru, and Venezuela. This study includes ownership-structure information on Argentinean and Mexican firms directly from Reuters for the years 2005 and 2006.

The initial sample has 1168 Latin American firms after excluding firms in the financial sector. We use consolidated financial statements in U.S. dollars as reported by Economatica. Firms with fewer than three years of balance sheet data and firms with less than \$250,000 of total assets are dropped. State-owned firms, firms that do not have direct operations and that are used only as vehicles for other investments, and firms with leverage ratios lower than zero or higher than one are also excluded. Hadi's (1992, 1994) outliers method is also applied to the control variables.

The final sample has 806 Latin American firms from seven Latin American countries: Argentina, Brazil, Chile, Colombia, Mexico, Peru, and Venezuela. The sample is from 1996 to 2005 and covers an unbalanced panel of 6766 firm-year observations. To control for potential survivorship bias, a subset of firms was constructed with all firms present for the entire sample period. The main results were similar to those reported here.

2.1. Variables

2.1.1. Leverage

Following Rajan and Zingales (1995) we define leverage as the ratio of financial debt to debt plus equity. This study considers both book value and market-value equity. We first calculate **leverage1_bv** as the ratio of total debt financing (short- and long-term) to total debt financing plus the book value of equity.

Depending on the country, Economatica excludes commercial papers and bonds from financial debt, instead calling them “negotiable obligations”. Consequently, we use **leverage2_bv** as the main measure of leverage; measured as the ratio of total financial debt plus total negotiable obligations to total financial debt and negotiable obligations plus the book value of equity.

Our third leverage measure, **leverage2_mv**, is computed as **leverage2_bv** but uses market value of equity (defined as the number of shares outstanding times the stock price) instead of book value equity.

Alternative measures of leverage were also considered, such as total financial debt to total book value of assets, and total liabilities to

total assets. The results are robust with the use of these alternative measures.

2.1.2. Ownership

To measure a firm's ownership concentration, the Herfindahl index of the firm's ownership structure (**herf_ind**) is used. **herf_ind** is calculated as the sum of the squares of the fractions of equity held by each individual shareholder (as reported by Economática).

This index tends to be underestimated given that some shareholders could be related (i.e., have family ties). Grouping the ownership concentration if shareholder *i* has the same last name as shareholder *j* does not change the main results. Moreover, the sample includes just the main 10 shareholders, so the Herfindahl index has a downward bias. However, the sample includes more than 70% of the ownership ($\sum si > 0.70$) for more than 2/3 of the firms included. This situation, if anything, works in the study's favor because it tends to bias the *t*-values toward zero.

High levels of **herf_ind** represent high ownership concentration. If the firm has, for instance, only two shareholders who each own 50% of the company, the **herf_ind** will be 0.5 ($=sq(0.5) + sq(0.5)$). On the other hand, if the firm has 5 shareholders with 20% ownership each, the **herf_ind** will be 0.2, indicating less ownership concentration.

Jensen and Meckling (1976) predict that shareholders of a firm with more concentrated ownership may prefer less debt if debt brings more monitoring (see also Leland and Pyle, 1977; Diamond, 1984).

We argue that firm control plays an important role in Latin America given the lack of protection for minority shareholders. Firms will prefer debt over equity if issuing equity means sharing or losing control. Thus, the relation between leverage and ownership concentration is expected to be positive, and the more concentrated the firm's ownership is, the more likely current shareholders are to issue debt instead of equity when the company needs funds. That is, they will follow a debt over equity pecking order. This study argues that this effect should be more important or visible in Latin America than in the U.S. because ownership concentration, the percentage of family businesses, and the asymmetry of information in Latin America tend to be higher. In Latin America the relevant asymmetric-information problem is not between management and shareholders as it is in developed economies. Rather it is between large shareholders and the rest of the financing providers (small shareholders and debtholders). The reason is that controlling shareholders manage the "information flow" among all the parties involved. For a recent review of asymmetric information in Latin America, see Chong and López de Silanes (2007). Shareholders of firms with highly concentrated ownership will not want to share or lose firm control, and they will avoid issuing equity if they are confident about their firms' future.

On the other hand, if the firm's ownership structure is dispersed (that is, the firm has a low ownership concentration), the value of ownership control may lose importance for shareholders. In that case, other factors may play a role in making capital-structure decisions, leaving us without a clear direction for this relation. Du and Dai (2005) show empirical evidence from East Asian firms that supports the idea that controlling shareholders in companies with relatively small ownership concentration (high separation of cash flow rights and control rights) tend to increase leverage in order to retain control.

As argued before, several country-specific reasons explain why Latin American companies should be less leveraged than their U.S. counterparts. For example, debt markets in Latin American are smaller and less efficient. Fewer debt options, such as long-term bonds, are available to firms, and the cost of debt is high for the average firm in the region. For the majority of Latin American firms, financing is restricted to the banking system. Some firms can raise financing through multilateral agencies; however, these resources are restricted to very limited uses and the amounts provided are relatively low. Other bigger firms raise capital through the local bond market, and although this has increased, it is still smaller than the bond markets in

developed economies. The average interest rate, measured as the ratio of total financial expenses to total financial debt, was never below 20% in the 1996–2005 period. Thus, Latin American companies should be less leveraged than those in the United States. However, as shown in the next section, this is not the case. This study argues that firms in Latin America are more leveraged than they should be and that this overleverage can be explained by the firms' ownership structures.

2.1.3. Size

We use the log of sales as a measure of firm size (**log_sales**). A positive relation between leverage and size is expected. Bigger firms are less likely to default on their debt obligations because they are more diversified and because renegotiating and issuing long-term debt securities are cheaper alternatives. The log of book value of assets is considered a robustness check.

2.1.4. Taxes

We use the effective tax rate (**tax_rate**), which is calculated as the ratio of paid taxes to earnings before taxes. The paid taxes are calculated by subtracting earnings after taxes from earnings before taxes. Values for this variable are between zero and one.

The expected sign for this variable depends on the capital-structure theory used. On one hand, a positive sign is consistent with the trade-off theory (De Angelo and Masulis 1980; Graham, 1996a,b); on the other hand, a negative sign could be a proxy of firm's profitability and, consistent with the pecking-order theory, lower debt level will be expected (Myers, 1977; Myers and Majluf, 1984).

2.1.5. Asset tangibility

We use the lag value of the ratio of fixed assets to total assets to assess the firm's asset tangibility (**txa_ta**), which is a proxy for collateral value. The greater the collateral value of a firm's assets, the more value the debtholder can recover in case of default (Jensen and Meckling, 1976). A positive coefficient relating leverage and tangibility is expected.

2.1.6. Profitability

To assess the firm's profitability, we use the lag value of the ratio of earnings before interest and taxes to total assets (**ebit_ta**). The expected sign for this variable depends also on the capital-structure theory used. On one hand, a positive sign is consistent with the trade-off theory, which implies that the higher the firm's profitability, the higher the potential tax shields and therefore the higher the firm's debt level (De Angelo and Masulis, 1980); on the other hand, a negative sign indicates that the firm will finance its operations and investment opportunities using its internally generated cash flow instead of using debt (Myers and Majluf, 1984).

2.1.7. Growth opportunities

We consider two measures of firm's growth opportunities, the market-to-book ratio, **mve_bve**, and the ratio of capital expenditures to fixed assets (**capex_ta**). Capital expenditure is estimated as the one-year variation in fixed assets.

The expected sign of this variable is negative given the agency relation between managers and debtholders and given that growth opportunities are noncollateral assets; however, this relation depends on debt maturity because some of the agency problems could be mitigated issuing short-term debt (Myers, 1977). As in Titman and Wessels (1988), a negative relation between long-term debt and growth opportunities and a positive relation between short-term debt and growth opportunities are expected. In Latin America most debt obligations are short-term, therefore the study expects a positive relation between the leverage and the study's proxies for growth opportunities. La Porta et al. (1997) argues that the quality of legal protection has a significant effect on the ability of firms in different countries to raise external finance. Moreover, these authors find that

French Civil Law countries have particularly small, narrow debt and equity markets.

In addition, if this study argues that Latin American firms avoid issuing equity because their shareholders are interested in preserving their control, leverage ratios should be positively related to growth opportunities.

2.2. Descriptive statistics

2.2.1. Leverage

Table 1 presents the descriptive statistics for three main leverage measures across the sample and across countries for the period 1996–2005. As a whole, and using book values, the region has a **leverage1_bv** of 29.45%, and 32.16% when **leverage2_bv** (debt plus negotiable obligations) is used. These ratios are similar to the 34% reported in Rajan and Zingales (1995) for the U.S. for a 1989–1996 sample period, and are higher than the 24.85% reported in Flannery and Rangan (2006), which uses a more recent and comprehensive data set of U.S. firms from 1965 to 2001. These results show, as argued before, that Latin American firms are overleveraged compared with their U.S. counterparts.

In another paper, Booth et al. (2001) reported average leverage ratios for Brazil and Mexico of 30.3% and 34.7%, respectively, for the period 1985–1991. Table 1 reports similar leverage ratios for Brazil and Mexico—37.78% and 31.96%, respectively—for the period 1996–2005.

Colombia and Venezuela have lower average book-value debt ratios in the sample, with **leverage2_bv** of 16.23% and 14.82%, respectively. However, they also have the fewest firm-year observations, making possible the idea that the leverage in these two countries is underestimated. After taking out Colombia and Venezuela, the average leverage in the sample is around 30% (**leverage1_bv**) and 32% (**leverage2_bv**).

In the whole sample, the number of observations falls dramatically when leverage using market values is considered (**leverage2_mv**). For the full sample, **leverage2_mv** averages 55.04%, which is much higher than similar leverage measures in the U.S. (Rajan and Zingales, 1995; Flannery and Rangan, 2006). Although this difference is consistent with the higher market-to-book ratios historically shown by U.S.

Table 1
Measures of leverage.

	Firm-year observations	leverage1_bv (%)	leverage2_bv (%)	leverage2_mv (%)
Argentina	716 (282)	29.83 (23.52)	34.36 (24.31)	39.24 (26.51)
Brazil	2822 (1242)	35.86 (25.29)	37.78 (25.80)	84.09 (32.40)
Chile	1363 (524)	20.84 (17.75)	24.35 (19.40)	30.38 (23.73)
Colombia	174 (24)	14.27 (17.20)	16.23 (18.06)	70.91 (39.13)
Mexico	1319 (622)	27.70 (23.48)	31.96 (24.25)	29.76 (26.08)
Peru	561 (83)	32.01 (22.27)	32.01 (22.27)	33.32 (24.46)
Venezuela	212 (83)	14.82 (16.14)	14.82 (16.14)	36.37 (30.71)
Full sample	7167 (2,860)	29.45 (23.77)	32.16 (24.38)	55.04 (38.73)
Rajan and Zingales (1995): U.S. firms	2,580		34.00	24.00
Flannery and Rangan (2006): U.S. firms	111,106		24.85	27.83

leverage1_bv is defined as the ratio of total debt financing (short- and long-term) to total debt financing plus the book value of equity; **leverage2_bv** as the ratio of total debt financing (short- and long-term and negotiable obligations, which are debt instruments such as commercial paper and bonds outstanding) to total debt financing plus the book value of equity; **leverage2_mv** as the ratio of total debt financing (short- and long-term and negotiable obligations) to total debt financing plus the market value of equity. The firm-year observations column represents book value measures and market value measures (in parenthesis). The percentage represents mean values and standard deviation (in parenthesis) across the sample for the period 1996–2005. The last two rows show the leverage ratios for U.S. firms according to Rajan and Zingales (1995) for the period 1987–1991 and Flannery and Rangan (2006). The number of observations for Rajan and Zingales data is cross-sectional for one year (1991), and the number of observations for Flannery and Rangan (2006) represents 12,919 U.S. firms from 1965 to 2001.

Table 2
Summary statistics.

	log_sales	tax_rate	fxa_ta	ebit_ta	mve_bve	capex_ta	herf_ind
Argentina	11.55	0.26	0.51	0.05	0.89	0.03	0.26
	1.75	0.22	0.24	0.09	0.65	0.07	0.29
	711	201	716	716	274	688	36
Brazil	12.08	0.25	0.43	0.06	0.12	0.04	0.31
	1.85	0.17	0.23	0.09	0.35	0.08	0.27
	2775	999	2822	2822	1194	2660	2188
Chile	11.16	0.16	0.54	0.07	1.20	0.04	0.35
	2.02	0.13	0.24	0.07	0.84	0.07	0.25
	1324	462	1363	1363	514	1341	1031
Colombia	11.05	0.25	0.30	0.04	0.20	0.03	0.24
	2.01	0.18	0.17	0.07	0.35	0.05	0.20
	174	29	174	174	30	164	112
Mexico	12.41	0.28	0.48	0.07	1.24	0.05	0.29
	2.01	0.17	0.23	0.08	0.91	0.07	0.26
	1312	453	1319	1319	545	1274	56
Peru	10.48	0.19	0.53	0.06	0.52	0.04	0.41
	1.38	0.17	0.22	0.08	0.43	0.07	0.30
	559	63	561	561	70	532	449
Venezuela	11.04	0.21	0.54	0.05	0.36	0.06	0.34
	1.93	0.17	0.23	0.08	0.38	0.12	0.24
	210	48	212	212	70	198	80
Full sample	11.73	0.24	0.48	0.06	0.65	0.04	0.33
	1.96	0.17	0.24	0.08	0.82	0.07	0.26
	7065	2255	7167	7167	2697	6857	3952

This table presents summary statistics of the proxies for size (**log_sales**), taxes (**tax_rate**), tangibility (**fxa_ta**), profitability (**ebit_ta**), growth opportunities (**mve_bve** and **capex_ta**), and ownership concentration (**herf_ind**). The first and second number are the mean values and the standard deviation across the sample for the period 1996–2005, respectively, and the third number is the number of firm-year observations.

firms, market liquidity and high transaction costs could also considerably affect market values in the sample. See Chong and López-de-Silanes (2007) for a recent account of the development of Latin American financial markets.

2.2.2. Debt determinants

Table 2 presents the descriptive statistics for the seven main debt determinants measured across the sample and across countries for the period 1996–2005. The descriptive statistics of the sample are consistent with those reported by Booth et al. (2001) for Mexico and Brazil. According to the sample, Latin American firms present an average effective tax rate of 24%, an average proportion of fixed assets of 48%, an average EBIT over total assets of 6%, an average market-to-book of 0.65, and an average ratio of capital expenditure to total assets equal to 4%.

As expected, and consistent with La Porta et al. (1997) and more recently Chong and López-de-Silanes (2007), the Latin American firms in the sample tend to be highly concentrated, with an average Herfindahl index of 0.33 for ownership. Peruvian firms show the highest concentration, with a Herfindahl index of 0.41, and Colombia the lowest with 0.24. Chong and López-de-Silanes (2007) reported the following ownership-concentration levels for the countries in their sample (except Peru): Argentina, 53%; Brazil, 57%; Chile, 45%; Colombia, 63%; Mexico, 64%; and Venezuela, 51%.

3. Results

In this section we first estimate leverage using as capital-structure determinants the proxies for growth opportunities, asset tangibility, size, and profitability. Then we include proxies for ownership concentration to study the effect of ownership control on leverage. The estimations consider fixed-effect models. The study estimates separate regressions including industry dummies, following Economica industry classification instead of firm dummies, and with no differences in the results. The findings are also robust when lagged

Table 3
Determinants of leverage.

Panel A: dependent variable: leverage2_bv								
capex_ta	0.11*** (2.89)	0 (0.02)	0.13* (1.82)	0.20** (2.25)	−0.23 (0.67)	−0.09 (0.69)	0.27** (2.18)	−0.02 (0.14)
fxa_ta	0.05*** (4.15)	0.08** (1.98)	0.06** (2.29)	0.11*** (4.67)	0.16 (1.48)	0.08** (2.43)	−0.08* (1.80)	−0.08 (1.10)
log_sales	0.03*** (18.89)	0.05*** (8.89)	0.03*** (7.11)	0.04*** (12.93)	0.03*** (4.90)	0.03*** (8.67)	0.02*** (3.04)	0.03*** (3.71)
ebit_ta	−0.62*** (17.39)	−0.85*** (7.51)	−0.48*** (6.88)	−0.72*** (10.44)	−0.39* (1.73)	−1.04*** (11.02)	−0.55*** (4.30)	−0.42** (2.45)
Observations	6766	684	2620	1304	164	1267	531	196
Country	All	AR	BR	CL	CO	MX	PE	VE
R ²	0.08	0.17	0.06	0.21	0.15	0.13	0.07	0.13
Fixed-effects	country	—	—	—	—	—	—	—
Dummies	Year	Year	Year	Year	Year	Year	Year	Year
Panel B: dependent variable: leverage2_mv								
capex_ta	0.21*** (2.89)	−0.2 (0.82)	0.27* (1.96)	0.09 (0.81)	3.04 (1.38)	−0.12 (0.51)	0.35 (0.79)	0.17 (0.43)
fxa_ta	−0.03 (1.28)	0.34*** (4.90)	−0.32*** (6.28)	0.21*** (4.79)	−0.11 (0.15)	0.15*** (2.95)	0.01 (0.05)	0.17 (0.64)
log_sales	0.04*** (11.00)	0.01 (0.84)	0.06*** (8.30)	0.03*** (5.16)	0.16*** (3.17)	0.02*** (2.77)	0.02 (1.09)	0.04 (1.44)
ebit_ta	−0.56*** (8.01)	−0.74*** (4.41)	−0.14 (1.19)	−1.43*** (8.24)	−1.22 (0.76)	−1.36*** (9.20)	−0.77** (2.01)	−0.17 (0.29)
Observations	2780	278	1189	513	22	618	79	81
Country	All	AR	BR	CL	CO	MX	PE	VE
R ²	0.07	0.25	0.14	0.29	0.68	0.16	0.14	0.17
Fixed-effects	country	—	—	—	—	—	—	—
Dummies	Year	Year	Year	Year	Year	Year	Year	Year

This table presents OLS regression using panel data with country fixed effects and OLS regression for each country using fixed effect. The dependent variable is **leverage_bv** in Panel A and **leverage_mv** in Panel B. The independent variables are defined in Table 2. The regression includes an intercept whose coefficient is not reported. Robust *t*-values are in parenthesis. * = significant at 10%; ** = significant at 5%; *** = significant at 1%.

values instead of current values of the independent variables are used in the estimation.

In Table 3, Panel A, **leverage2_bv** is the dependent variable (similar nonreported results are obtained using **leverage1_bv**). Consistent with Titman and Wessels (1988), Rajan and Zingales (1995), and Booth et al. (2001), this study finds that asset tangibility (**fxa_ta**) and size (**log_sales**) influence positively firms' leverage in the sample of Latin American firms. Both coefficients are statistically significant for the full sample and remain positive and significant for each country. In the case of tangibility, the coefficient is negative only in Peru.

The effect of profitability (**ebit_ta**) is negative and significant in each country, consistent with previous evidence. The effect of the proxy for growth opportunities (**capex_ta**) is positive and significant in the full sample regression and positive and significant in Brazil, Chile, and Peru. Although inconsistent with the international evidence (Rajan and Zingales, 1995), the positive relation this study finds between leverage and growth opportunities in Latin America can be explained by the desire of Latin American firms to avoid equity issuances and the consequent loss of control. Another reason is the higher use of short-term debt in the region (Titman and Wessels, 1988).

The results remain when estimating the regression model using **leverage_mv** as the dependent variable. However, as shown in Panel B, the number of observations for this estimation fell significantly.

In sum, this study concludes that leverage is positively related to asset tangibility and size, and negatively related to profitability in Latin American companies. A positive relation between leverage and growth opportunities is also found, supporting the idea that debt is preferred to equity in this region. Other reasons could also explain this preference, such as the high transaction costs and low liquidity typical in the Latin American financial markets (Chong and López-de-Silanes, 2007).

Table 4 includes in the model the study's tax proxy (**tax_rate**), and more important, the ownership-concentration variable (**herf_ind**). These regressions are run only on data for 2005 because the data on

ownership concentration for Argentina and Mexico, collected from Reuters, is available only for that year. The study runs the same regressions, excluding Argentina and Mexico, with a panel of 1996–2005 and 2001–2005, obtaining the same results. The results from Table 3 regarding the positive relation between leverage and growth opportunities, asset tangibility, and size, and the negative relation between leverage and profitability continue to be strong in Table 4. Table 4 also reports a negative and significant relation between leverage and taxes, consistent with the pecking-order theory and the idea that tax rate is a proxy for firm profitability.

The first column of Table 4 reports a negative but not significant coefficient for ownership concentration. This result can be explained by the agency theory of capital structure (Jensen and Meckling, 1976; Fama 1980; Fama and Jensen, 1983), which argues that high levels of ownership concentration induce firms to issue less debt to avoid monitoring. Although this negative relation is consistent with the agency theory literature, the relation is not entirely applicable to the study's data set given that top managers in Latin America usually have very close relationships with main shareholders. Another possibility is that when control is not an issue, undiversified shareholders exert their aversion to debt.

Table 4 column 2 includes in the estimation the square value of **herf_ind** to test whether the relation between leverage and ownership exhibits a nonlinear relation. This study obtains a negative and significant coefficient for **herf_ind** and a positive and significant coefficient for **herf_ind**². This result is interpreted as evidence of a U-shape relation between leverage and ownership, where firms with low ownership concentration exhibit a negative relation with leverage, and firms with high ownership concentration exhibit a positive relation with leverage.

Table 4 columns 3 through 5 follow Morck et al. (1988) and split the **herf_ind** variable in two. This study considers three inflection points: 0.4, 0.5, and 0.6. **herf_ind** is then split into **herf_ind**_{0–x} and **herf_ind**_{x–1}. **herf_ind**_{0–x} is equal to **herf_ind** if **herf_ind** is lower than *x* (inflection point), and **herf_ind**_{0–x} is equal to *x* otherwise. On the

Table 4
Determinants of leverage with ownership.

Dep. variable	(1) leverage2_bv	(2)	(3)	(4)	(5)	(6)	(7) leverage2_mv
Panel A: regressions for all countries year 2005							
capex_ta	−0.14 (0.81)	−0.11 (0.66)	−0.11 (0.63)	−0.11 (0.67)	−0.12 (0.71)	0.00 (0.03)	0.10 (0.61)
fxa_ta	0.11** (1.97)	0.12** (2.26)	0.12** (2.20)	0.12** (2.23)	0.12** (2.32)	0.11** (2.02)	0.06 (1.17)
log_sales	0.03*** (4.94)	0.03*** (4.70)	0.03*** (4.76)	0.03*** (4.78)	0.03*** (4.80)	0.02*** (4.15)	0.01*** (2.61)
ebit_ta	−0.28** (1.98)	−0.29** (2.08)	−0.29** (2.11)	−0.29** (2.10)	−0.29** (2.08)	−0.34** (2.40)	−0.21 (1.62)
tax_rate	−0.28*** (4.30)	−0.27*** (4.18)	−0.26*** (4.10)	−0.27*** (4.19)	−0.27*** (4.26)		−0.23*** (3.81)
herf_ind	−0.02 (0.58)	−0.46*** (3.33)				−0.50*** (3.54)	−0.42*** (3.22)
herf_ind²		0.47*** (3.30)				0.50*** (3.44)	0.44*** (3.22)
herf_ind_{0−x}			−0.28*** (2.99)	−0.23*** (2.97)	−0.20*** (2.99)		
herf_ind_{x−1}			0.18** (2.30)	0.27*** (2.64)	0.41*** (3.01)		
Inflection (x)			0.40	0.50	0.60		
Observations	403	403	403	403	403	403	403
Country	All	All	All	All	All	All	All
R ²	0.10	0.13	0.13	0.13	0.13	0.09	0.09
Panel B: Individual country regressions							
Dep. variable	leverage2_bv						
capex_ta	−0.30 (0.38)	0.24** (1.99)	0.15 (1.02)	0.03 (0.07)	−0.64 (1.15)	0.47 (1.31)	0.49*** (4.22)
fxa_ta	−0.01 (0.04)	0.19*** (4.59)	0.07** (2.05)	0.05 (0.40)	−0.01 (0.08)	0.03 (0.37)	0.10 (0.85)
log_sales	0.11*** (4.47)	0.02*** (3.39)	0.04*** (14.11)	0.03** (2.27)	0.06** (2.73)	0.05*** (4.12)	0.01 (0.21)
ebit_ta	−1.69*** (5.28)	−0.20** (2.12)	−0.72*** (6.61)	−0.42 (1.03)	−0.21 (0.44)	−0.61*** (2.72)	−0.06 (0.42)
tax_rate	−0.36** (2.19)	−0.28*** (6.32)	0.07 (0.98)	−0.09 (0.58)	−0.02 (0.11)	−0.17 (1.47)	−0.05 (0.72)
herf_ind	−0.41 (1.18)	−0.35*** (3.22)	−0.20** (1.96)	−0.15 (0.60)	−0.09 (0.20)	−0.04 (0.18)	−0.32 (0.72)
herf_ind²	0.25 (0.68)	0.38*** (3.32)	0.13* (1.92)	0.39 (1.10)	−0.04 (0.08)	−0.02 (0.07)	0.24 (0.58)
Observations	21	1027	556	82	40	155	43
R ²	0.70	0.11	0.30	0.18	0.22	0.10	0.29
Country	AR	BR	CL	CO	ME	PE	VE
Fixed-effects	–	–	–	–	–	–	–
Dummies	–	Year	Year	Year	–	Year	Year

This table presents the OLS regression for all countries in the year 2005 (Panel A) and OLS regression using fixed-effect panel data (2001–2005) for each country except Argentina and Mexico, where OLS results are reported for the year 2005. The dependent variable is **leverage2_bv** for columns 1 through 6 and **leverage2_mv** for column 7 in Panel A. Individual country regressions use **leverage2_bv** as the dependent variable in Panel B. The independent variables are defined in Table 2. The regression includes an intercept whose coefficient is not reported. Robust *t*-values are in parenthesis. * = significant at 10%; ** = significant at 5%; *** = significant at 1%.

other hand, **herf_ind_{x−1}** is equal to *x* (inflection point) if **herf_ind** is lower than *x*, and **herf_ind_{x−1}** is equal to **herf_ind** otherwise.

In all the study's regressions the coefficient for **herf_ind_{0−x}** is negative and significant, and the coefficient is positive and significant for **herf_ind_{x−1}**. In sum, these results show a U-shape relation between leverage and ownership concentration. This U-shape relation is consistent with the argument that, as the firm's ownership concentration increases, firms avoid issuing equity because their shareholders do not want to lose or share their control rights.

Consistent with this idea, Table 3 (panel A) shows a positive and significant coefficient for the proxy of growth opportunities (**capex_ta**), which demonstrates that in the sample of Latin American firms' growth, is mainly financed by debt. This effect is still positive, although not significant, when ownership concentration is included in the study's regressions. When the same regression analysis (not reported, but available from the authors) is performed using panel data fixed-effect equations, excluding Argentina and Mexico where ownership data was available only for the year 2005, **capex_ta** was

positive and significant and the U-Shape relation between leverage and ownership concentration was still present.

The country regressions in Panel B of Table 4 also exhibit the U-shape relation, although this relation is statistically significant only in Brazil and Chile, the two countries with more observations in the sample. With the exception of Mexico, the coefficient signs, although not significant, are correct for all other countries. This is remarkable given the small number of observations for countries such as Argentina (21), Colombia (82), and Venezuela (43), which reduces considerably the statistical power of the estimation.

We perform several robustness checks (results are not reported) to test our main result on the relationship between leverage and ownership concentration. First we include control variables for macro-economic effects such as stock market capitalization as a percentage of GDP, economic growth as a percentage change in GDP, and inflation rate. Second, we consider controls for trading volume and stock liquidity. We also repeat the regressions in Tables 3 and 4 for 1996–2005, 2001–2005, 1996–2000, and 2005 sample periods. The main

result of a U-shape relation between leverage and ownership concentration remains.

4. Conclusions

This paper uses a comprehensive database of 806 nonfinancial Latin American firms from 1996 to 2005 to be the first to test how ownership concentration affects capital-structure decisions in Latin America. Latin American markets present us with an ideal setting to study the effect of ownership concentration on leverage given the high concentration of ownership in Latin American firms. At the aggregate level the results support the idea that at high levels of ownership concentration, the relation between leverage and ownership is positive, consistent with the argument that companies with highly concentrated ownership do not seek equity financing in order to avoid losing control. These results are consistent across countries, although they are strong only for Brazil and Chile, the two countries with the largest number of observations in the sample. Consistent with the study's control argument, this study reports that Latin American firms with more growth opportunities tend to exhibit higher leverage.

This study also provides empirical support for previous findings that size, growth opportunities, asset tangibility, profitability, and taxes are statistically and economically important in explaining the leverage in the sample of Latin American firms. Firms that are larger, have high-growth opportunities, and are less profitable tend to have more debt.

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